

intelligently.

We cannot truly plan, because we do not understand the future—but this is not necessarily bad news. We could plan *while bearing in mind such limitations*. It just takes guts.

The Beauty of Technology: Excel Spreadsheets

In the not too distant past, say the precomputer days, projections remained vague and qualitative, one had to make a mental effort to keep track of them, and it was a strain to push scenarios into the future. It took pencils, erasers, reams of paper, and huge wastebaskets to engage in the activity. Add to that an accountant's love for tedious, slow work. The activity of projecting, in short, was effortful, undesirable, and marred with self-doubt.

But things changed with the intrusion of the spreadsheet. When you put an Excel spreadsheet into computer-literate hands you get a “sales projection” effortlessly extending ad infinitum! Once on a page or on a computer screen, or, worse, in a PowerPoint presentation, the projection takes on a life of its own, losing its vagueness and abstraction and becoming what philosophers call reified, invested with concreteness; it takes on a new life as a tangible object.

My friend Brian Hinchcliffe suggested the following idea when we were both sweating at the local gym. Perhaps the ease with which one can project into the future by dragging cells in these spreadsheet programs is responsible for the armies of forecasters confidently producing longer-term forecasts (all the while tunneling on their assumptions). We have become worse planners than the Soviet Russians thanks to these potent computer programs given to those who are incapable of handling their knowledge. Like most commodity traders, Brian is a man of incisive and sometimes brutally painful realism.

A classical mental mechanism, called anchoring, seems to be at work here. You lower your anxiety about uncertainty by producing a number, then you “anchor” on it, like an object to hold on to in the middle of a vacuum. This anchoring mechanism was discovered by the fathers of the psychology of uncertainty, Danny Kahneman and Amos Tversky, early in their heuristics and biases project. It operates as follows. Kahneman and Tversky had their subjects spin a wheel of fortune. The subjects first looked at the number on the wheel, *which they knew was random*, then they were asked to estimate the number of African countries in the United Nations. Those who had a low number on the wheel estimated a low number of African nations; those with a high number produced a higher estimate.

Similarly, ask someone to provide you with the last four digits of his social